of which were under special protection. Significant amounts of new finds were recovered from eight sites and data on the extents of six was updated.

The research project ran under the name of OpenHeritage (a project supported by the Horizon2020 scheme), and it was conducted within the frames of the CEU Cultural Heritage Studies programme. It studied how local communities can be involved in protecting cultural heritage by developing the cultural heritage site at Pomáz– Nagykovácsi-puszta for local education and cultural tourism purposes. One of the aims was to understand the perspectives of civilians regarding the importance of archaeological sites. On every occasion open lectures by leading experts of the relevant archaeological period were organised for volunteers (Fig.43), and after the lectures field surveys were conducted, where new information on the condition and scientific potential of the sites was gained. The detailed descriptions were in fact assessments of the cultural values of different locations. The core activities were archaeological fieldwork and metal detector-aided survey as well as droned-based aerial photography and high-precision (RTK) geodetic surveys when necessary. Photo documentation, accurate description on the state of locations was completed.

Assessing significance – methodological considerations

While preparing the assessment several aspects needed to be considered. A system of criteria suitable for practical application and valid for different kinds of sites needed to be created, where the differences could be expressed in a measurable form. These needed to include all scientific aspects possible and besides that, the special aspects of local communities and volunteers. Furthermore, the study also took into consideration social values and benefits in the context of archaeological site evaluation. Setting the criteria and their application is of course subjective, depending on the experience and knowledge of the surveyor and always valid in a particular relational system. It is desirable for the surveys to be conducted by a team, discussing the issues raised and summarising the views of the participants.

The evaluation system used in the project was conceived in the form of a spreadsheet, containing a wide range of criteria, grouped into 5 larger units. The individual characteristics were marked with points ranging from 0 to 3 in accordance with the lower or higher degree of validity regarding the site. The more data available on the site the more accurate the assessment will be. From experience during the pilot project, it is highly recommended to complete the assessment while visiting the site. It requires significant preparation before the site visit, as the assessment takes also into consideration the previous archaeological research. If there had been previous field research projects, excavations, or field surveys, one can make the assessment with high precision.

The first part of the evaluation estimated the scientific significance of the site. Standard categories used in archaeological research and heritage assessment were listed: general characteristics of the site, like the level of preservation, rarity and uniqueness of the site, complexity of archaeological phenomena. Particular attention was paid to the research potential of the site i.e. the importance of the site for future investigations based on the recovered archaeological finds and features, the available academic results and scholarly interpretations, and the present condition of the archaeological heritage at the site. The assessment continued with the individual features and finally concentrates on the finds.

The second unit dealt with the cultural heritage significance. Here broader heritage aspects were listed, like historical narratives or connection to natural heritage values which have the capacity to enhance the value of a site. Elements of built heritage for example have a great and lasting impact on the appearance of the site. Mentions of an archaeological site in literary works (poetry, drama) relate to a special status, values which are obvious for non-archaeologists also. These aspects are particularly important in the case of places and sites which can be associated with great historical events, like a battle, or are connected to great historical figures. Additional points were given for the presence of religious locations, churches for example.

The next unit displayed key aspects for the protection of the site, forming a summary of a risk assessment expressed in points, including short- and long-term factors, and among them human and natural causes. An inverse evaluation was applied: the more endangered is the site, the more points it gets, bearing in mind that the appraisal might contribute to its protection. Volunteers and local communities could add several new aspects to the values of cultural heritage locations,

Figure 44: Bronze Age sword. Pomáz, Kő-hill. (Rácz, T.)

which must also be considered during the assessment of sites. Local awareness of the site, local historical events, legends, memorials, statues represent non-scientific considerations, but create living connections between a community and a historical location.

The last section explored the possibilities of 'valorisation': income generating possibilities, like tourism, or social benefits with particular emphasis on educational aspects. This involves the interpretation of the site. Furthermore, conditions and infrastructural aspects of a visit were also taken into consideration, including recreational possibilities, facilities and infrastructure (toilets). Accessibility and available information at the site on the archaeological heritage meant additional points for the site.

Description of the sites

Four archaeological sites of different ages and of very different character (fortification, settlement, church, and a hoard) were selected for the survey, to demonstrate the complexity of the assessment process, and they were also used as test cases to see the applicability of the spreadsheets used in the pilot project.

The first site was the fortified hilltop settlement at Pomáz, Kő-hill, which is an important prehistoric complex in the region with a long research history. The natural heritage of the site is its most attractive element, and it is a popular hiking place for visitors from Budapest. The archaeological features (earthworks, terraces) are parts of the natural landscape, but they are not presented for the wider public (no information display at the site). Social events are also organized here. An annual memorial event takes place each year on 15 March to commemorate the 1848 revolution. This is connected to the fact that a renowned nineteenth century poet, Sándor Petőfi, wrote a poem here, and he was also a key figure of the revolution and of the war of independence in 1848-49. During the survey a great quantity of Bronze Age pottery and one of the best finds of the whole campaign, a Bronze Age sword, was found here (Fig.44). The project also contributed to a survey of the site which extended the boundary of the archaeological features.

Figure 45: Finds of a metal detector-aided survey at Budakalász, Dunai Kisföldek (Havas, V.)

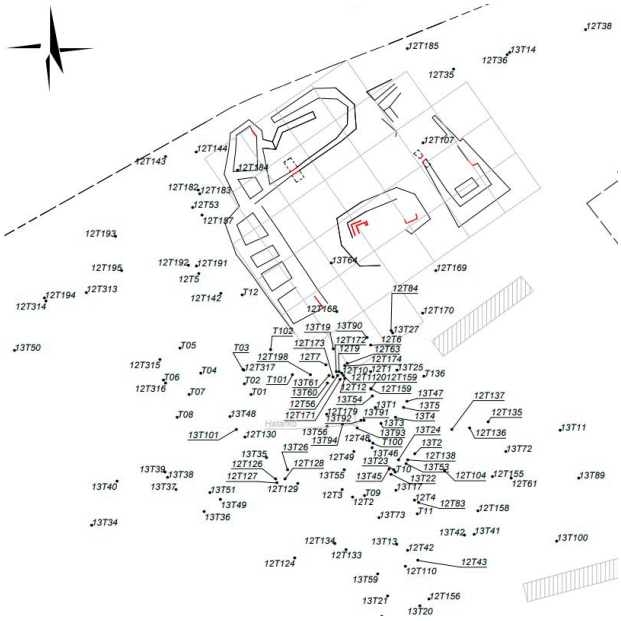

Figure 46: The remains of a medieval monastic manorial complex at Pomáz, Nagykovácsi-puszta, with the finds of a metal detector-aided survey. (Harmath, A.)

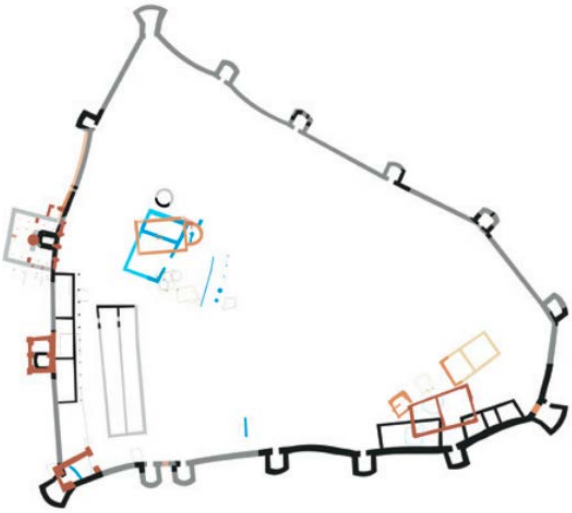

Figure 47: Visegrád, Sibrik-hill. Remains of the Roman and medieval fortification. (Buzás, G.)

The second example represents a depot from Budakalász, Dunai Kisföldek. This was discovered by chance in the context of the condition assessment of an adjacent site under special protection, when metal detector survey was carried out in a small grassland area. In 1954, only a fragment of a Roman milestone was found here, but surprisingly during the project, in addition to a few bronze coins, some 60 Roman brooches and brooch fragments were unearthed, as well as a Celtic coin (Fig.45). It is assumed that the brooches and other accessories refer to a hidden treasure trove or a Roman cemetery. This is a major change in knowledge of the site, and the academic research potential of the archaeological features below the surface is now much bigger, compared to previously.

The third site represents a different archaeological heritage and a complex research history. Ruins in the area of Pomáz, Nagykovácsi-puszta have been recorded since the nineteenth century. They were identified as medieval, but the function and the identification of the remains of a stone-built building complex was very much debated. During the last fifteen years complex investigations (field survey, remote sensing, landscape archaeology, environmental archaeology, building survey, excavation, etc.) were carried out at the site and in the area around the ruins. Parts of a complex water system, traces of agricultural fields, iron and glass production were identified. The church and the workshop buildings represent a manorial complex connected to a major Cistercian monastery nearby (Fig.46). Research was carried out in the framework of field schools, university educational projects and community archaeology programmes. The present pilot project was also connected to this site, as introductory lectures and educational programmes were organized in the buildings of the modern farm nearby.

Our fourth example, Visegrád, Sibrik hill is a key site for Hungarian archaeological research. It was a Roman fort, and further used by different populations later (Carolingians, Slavs) and rebuilt after the foundation of the Hungarian state. First it became a county centre and while displaying complex forms of settlement development, it was later a royal centre, but the visible remains are less attractive for the general audience. Recent archaeological investigations (geophysical survey, remote sensing, GIS, landscape archaeological studies, etc.) confirmed the exceptional research potential of the site (Fig.47).

Assessing significance – situation report

The spreadsheets (see tables 4-7) display a great variety of factors in the evaluation process. Pomáz, Kő-hill received average values in the general assessment. For the scientific significance we suggested 21 points, while the cultural heritage significance valued only 8 points. It is moderately exposed to destruction and

Figure 47: Archaeological trail at the Pomáz, Nagykovácsi-puszta site. Remains of the medieval water system. (Laszlovszky, J.)

looting (4 points). It is an important site for the research of a particular archaeological period. It is also highly important for the local community (9 points) but its importance cannot be compared to Visegrád, for example. It offers an excellent location for a complex (archaeological, geological, natural) educational trail with significant amenity values (12 points). The site scored a total of 54 points.

Significantly lower values came out in case of Budakalász, Dunai Kisföldek (a total of 32 points). Both its cultural heritage significance and importance for the local community was worth 0 points. It is an unknown site, without any visible features. The value of the site lies in its uniqueness, its potential for new scientific results and the types of artefacts. The area nearby holds considerable aesthetic values and recreational possibilities; thus, the character of the site can be changed. On the other hand, the danger of looting the site is relevant, as illegal metal detecting can destroy undetected archaeological finds and features.

The archaeological site Pomáz, Nagykovácsi-puszta has received a relatively high overall score: 69 points. Compared to the historical significance of the other medieval site in our pilot project (Visegrád, Sibrik-hill: 71 point) this high point of the Pomáz site can be seen as a surprise. The spreadsheet clearly shows that in this case the scientific importance combined with the research potential and the social benefits of the site made up together this high figure. The relatively unknown research topics (monastic manorial complexes, medieval glass production) make this site unique. The educational and community archaeology programmes also create a significant interest for this historical monument. Thus, for the assessment of importance this complex heritage practice should also be taken into consideration, which is also shown in the relatively high figures in cultural heritage importance and amenity values, 14-17 points respectively (Fig.47).

The scientific significance of Visegrád, Sibrik-hill received almost the maximum of points (37 points). Its cultural heritage significance is also above average (15 points). However, cultural heritage interpretation of the site is less developed, it does not show the significance of the site. These aspects are clearly visible in the low values in the social benefit and touristic attraction aspects (11 points). The site is well protected, continuously monitored and researched – it is not exposed to destruction or looting (1 point). It has the potential to enhance public understanding of history in case it would receive infrastructural development. We granted a total of 71 points to the site.

Conclusion

The process of bestowing archaeological sites with a protected status has a long history in Hungary. Between 2004 and 2016, numerous sites were declared protected with an institutional framework. Currently there is no elaborated methodology available to researchers for expressing the value of a site. In our study, we developed a new classification system.

Besides strict scientific approach, several new criteria needed to be added based on the results of our surveys, for example the perspectives of citizen science. A key aspect of our research project was the involvement of non-professionals. Our research has shown that the more information we can obtain about a site, for example through excavation or metal detecting, the more its value increases as its characteristics become identifiable. The disclosure of the values of sites buried, unexplored or unpublished helps scholars to deal with the site according to its real scientific importance and can strengthen the identity of the local communities concerned.

Archaeological sites have different scientific or cultural values. Therefore, their heritage management, research and protection require different procedures. Any decision can only be taken based on a responsible assessment of the site. The evaluation must be elaborated by experienced scholars intensively working on the respective archaeological sites. This interpretation needs to be harmonized with expectations and different value systems: the perspectives of museums, volunteers, local residents, infrastructural investors must also be included in a scientific assessment of cultural values. Therefore, the proposed scheme is an integration of such different perspectives.

The procedure can have various outcomes. Several sites need extra protection and attention because of imminent destruction. Treasure hunters and illegal metal detectorists had been known to raid archaeological sites prioritized by heritage protection. In other cases, archaeological sites had been endangered by unauthorized construction works by landowners and developers. Others can enhance tourism due to their potential and several locations can be used for educational or recreational purposes. The assessment can provide actionable data for protection and heritage utilization. This scheme can also be further developed by using international best practices and other assessment systems.

TABLE 4: Assessing the significance of the Pomáz, Kő-hill archaeological site

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Scientific significance	0	1	2	3
level of preservation of the site			x	
rarity and uniqueness of the site			x	
complexity of archaeological phenomena			x	
complexity of cultural layers and historical periods			x	
potential for new discoveries and scientific results			x	
complexity of scientific information		x		
importance revealed in published scientific literature or documentation		x		
special conditions (waterlogged, sunken or closed environment)		x		
significance of individual features		x		
representative of a given period or culture			x	
overall quality of archaeological finds			x	
quantity of archaeological objects		x		
diversity of archaeological finds and features			x	
treasure finds, hoards or rare types of artefacts	x			
Evaluation expressed in points		5	16	
Total	21			

Cultural heritage significance (archaeological, historical, architectural)	0	1	2	3
historical city centre or other monumental environment	x			
historical landscape features			x	
natural heritage (nature reserve, biodiversity, geological significance)			x	
association with important historical events	x			
association with important historical figures	x			
mentions in historical sources		x		
presence in historical illustrations		x		
visible built heritage, historic buildings	x			
mentions in literary works (poetry, novel, drama, etc.)			x	
features of art historical importance	x			
cultural practices of a particular society or civilization (intangible heritage)	x			
religious, ceremonial location, proof for ritual practices	x			
Evaluation expressed in points		2	6	
Total	8			

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Exposure to destruction and looting	**0**	**1**	**2**	**3**
danger of destruction by building operations		x		
danger of destruction by agricultural works	x			
exposure to illegal excavation, looting and vandalism		x		
exposure to illegal metal detecting		x		
erosion and other natural deterioration		x		
Evaluation expressed in points		4		
Total	4			

Importance for the local community	**0**	**1**	**2**	**3**
local awareness of the site				x
place of remembrance, location of a memorial or statue			x	
local historical events or associated legends			x	
local community events organised on-site			x	
Evaluation expressed in points			6	3
Total	9			

Amenity values (tourism, recreation and education)	**0**	**1**	**2**	**3**
accessibility			x	
visibility and aesthetic value			x	
connection to other sites or cultural/touristic routes		x		
recreational possibilities			x	
facilities and infrastructure			x	
potential to enhance public understanding of history		x		
potential to understand archaeological heritage		x		
valuable remains for educational purposes		x		
Evaluation expressed in points		4	8	
Total	12			

Overall collected points: 54.

TABLE 5: Assessing the significance of the Budakalász, Dunai Kisföldek archaeological site

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Scientific significance	0	1	2	3
level of preservation of the site	x			
rarity and uniqueness of the site				x
complexity of archaeological phenomena	x			
complexity of cultural layers and historical periods	x			
potential for new discoveries and scientific results				x
complexity of scientific information		x		
importance revealed in published scientific literature or documentation		x		
special conditions (waterlogged, sunken or closed environment)	x			
significance of individual features	x			
representative of a given period or culture			x	
overall quality of archaeological finds		x		
quantity of archaeological objects			x	
diversity of archaeological finds and features		x		
treasure finds, hoards or rare types of artefacts				x
Evaluation expressed in points		4	9	4
Total	17			

Cultural heritage significance (archaeological, historical, architectural)	0	1	2	3
historical city centre or other monumental environment	x			
historical landscape features	x			
natural heritage (nature reserve, biodiversity, geological significance)	x			
association with important historical events	x			
association with important historical figures	x			
mentions in historical sources	x			
presence in historical illustrations	x			
visible built heritage, historic buildings	x			
mentions in literary works (poetry, novel, drama, etc.)	x			
features of art historical importance	x			
cultural practices of a particular society or civilization (intangible heritage)	x			
religious, ceremonial location, proof for ritual practices	x			
Evaluation expressed in points				
Total	0			

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Exposure to destruction and looting	**0**	**1**	**2**	**3**
danger of destruction by building operations		x		
danger of destruction by agricultural works		x		
exposure to illegal excavation, looting and vandalism		x		
exposure to illegal metal detecting			x	
erosion and other natural deterioration	x			
Evaluation expressed in points		3	2	
Total	5			

Importance for the local community	**0**	**1**	**2**	**3**
local awareness of the site	x			
place of remembrance, location of a memorial or statue	x			
local historical events or associated legends	x			
local community events organised on-site	x			
Evaluation expressed in points				
Total	0			

Amenity values (tourism, recreation and education)	**0**	**1**	**2**	**3**
accessibility				x
visibility and aesthetic value			x	
connection to other sites or cultural/touristic routes			x	
recreational possibilities			x	
facilities and infrastructure	x			
potential to enhance public understanding of history	x			
potential to understand archaeological heritage		x		
valuable remains for educational purposes	x			
Evaluation expressed in points		1	6	3
Total	10			

Overall collected points: 32.

TABLE 6: Assessing the significance of the Pomáz, Nagykovácsi-puszta archaeological site

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Scientific significance	0	1	2	3
level of preservation of the site			x	
rarity and uniqueness of the site				x
complexity of archaeological phenomena				x
complexity of cultural layers and historical periods			x	
potential for new discoveries and scientific results			x	
complexity of scientific information				x
importance revealed in published scientific literature or documentation			x	
special conditions (waterlogged, sunken or closed environment)				x
significance of individual features			x	
representative of a given period or culture			x	
overall quality of archaeological finds			x	
quantity of archaeological objects				x
diversity of archaeological finds and features			x	
treasure finds, hoards or rare types of artefacts				x
Evaluation expressed in points			16	18
Total	34			

Cultural heritage significance (archaeological, historical, architectural)	0	1	2	3
historical city centre or other monumental environment	x			
historical landscape features			x	
natural heritage (nature reserve, biodiversity, geological significance)			x	
association with important historical events		x		
association with important historical figures	x			
mentions in historical sources		x		
presence in historical illustrations	x			
visible built heritage, historic buildings			x	
mentions in literary works (poetry, novel, drama, etc.)		x		
features of art historical importance		x		
cultural practices of a particular society or civilization (intangible heritage)			x	
religious, ceremonial location, proof for ritual practices			x	
Evaluation expressed in points		4	10	
Total	14			

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Exposure to destruction and looting	**0**	**1**	**2**	**3**
danger of destruction by building operations	x			
danger of destruction by agricultural works	x			
exposure to illegal excavation, looting and vandalism	x			
exposure to illegal metal detecting	x			
erosion and other natural deterioration		x		
Evaluation expressed in points		1		
Total	1			

Importance for the local community	**0**	**1**	**2**	**3**
local awareness of the site		x		
place of remembrance, location of a memorial or statue	x			
local historical events or associated legends	x			
local community events organised on-site			x	
Evaluation expressed in points		1	2	
Total	3			

Amenity values (tourism, recreation and education)	**0**	**1**	**2**	**3**
accessibility		x		
visibility and aesthetic value			x	
connection to other sites or cultural/touristic routes		x		
recreational possibilities			x	
facilities and infrastructure			x	
potential to enhance public understanding of history				x
potential to understand archaeological heritage				x
valuable remains for educational purposes				x
Evaluation expressed in points		2	6	9
Total	17			

Overall collected points: 69.

TABLE 7: Assessing the significance of the Visegrád, Sibrik-hill archaeological site

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Scientific significance	0	1	2	3
level of preservation of the site				x
rarity and uniqueness of the site				x
complexity of archaeological phenomena				x
complexity of cultural layers and historical periods				x
potential for new discoveries and scientific results				x
complexity of scientific information				x
importance revealed in published scientific literature or documentation				x
special conditions (waterlogged, sunken or closed environment)	x			
significance of individual features				x
representative of a given period or culture				x
overall quality of archaeological finds				x
quantity of archaeological objects			x	
diversity of archaeological finds and features				x
treasure finds, hoards or rare types of artefacts			x	
Evaluation expressed in points			4	33
Total	37			

Cultural heritage significance (archaeological, historical, architectural)	0	1	2	3
historical city centre or other monumental environment		x		
historical landscape features			x	
natural heritage (nature reserve, biodiversity, geological significance)			x	
association with important historical events			x	
association with important historical figures			x	
mentions in historical sources			x	
presence in historical illustrations		x		
visible built heritage, historic buildings		x		
mentions in literary works (poetry, novel, drama, etc.)	x			
features of art historical importance		x		
cultural practices of a particular society or civilization (intangible heritage)	x			
religious, ceremonial location, proof for ritual practices		x		
Evaluation expressed in points		5	10	
Total	15			

0= non-valid/ no data/ very low. **1**= low. **2**= average. **3**= high.

Exposure to destruction and looting	**0**	**1**	**2**	**3**
danger of destruction by building operations	x			
danger of destruction by agricultural works	x			
exposure to illegal excavation, looting and vandalism	x			
exposure to illegal metal detecting	x			
erosion and other natural deterioration		x		
Evaluation expressed in points		1		
Total	1			

Importance for the local community	**0**	**1**	**2**	**3**
local awareness of the site				x
place of remembrance, location of a memorial or statue				x
local historical events or associated legends	x			
local community events organised on-site		x		
Evaluation expressed in points		1		6
Total	7			

Amenity values (tourism, recreation and education)	**0**	**1**	**2**	**3**
accessibility			x	
visibility and aesthetic value			x	
connection to other sites or cultural/touristic routes				x
recreational possibilities		x		
facilities and infrastructure	x			
potential to enhance public understanding of history		x		
potential to understand archaeological heritage		x		
valuable remains for educational purposes		x		
Evaluation expressed in points		4	4	3
Total	11			

Overall collected points: 71.